17/8/26

EXPERIMENT 8

NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS(ARP, ICMP, TCP/UDP)

AIM: To desiover live hosts in a network using Nmap by preforming ARP, ICMP, TCP, UDP using host dislovery scans on TryHackMe platform.

ALGORITHM:

Step 1: Start TryHackMe machine and identify target IP address/range.

Step 2: Open Terminal -

Step 3: Preform ARP scan to dasiover hosts on local network.

Step 4: Preform ICMP echo scan.

Step 5: Preform a TCP SYN scan to discovre hosts responding to TCP.

Step 6: Preform a UDP scan to identify UDP ports.

Step 7: Observe and record IP address of live host.

Step 8: Compare the results obtained.

Step 9: Stop TryHackMe

PROGRAM:

1. ARP Host Discovery.

sudo nmap -PR <target-ip/range>.

[-PR. preforms ARP ping scan]

2. ICMP Host Discovery.

sudo nmap -PE <target-ip/range>

[-PE - sends ICMP Echo requests]

3. TCP SYN HOST Discovery.

sudo nmap -PS80, 443 <target-ip/range>

[-PS - option performs TCP SYN discovery on specified ports]

4. UDP HOST discovery.

Sudo nmap -PU 53, 161 <target -ip/range>.

5. Analyze results:

- Nmap displays the host that are detected as being up.
- Target IP address.
- Host status
  - Respons / latency information
  - Scanned ports.

RESULT:

Thus, NMAP was succesfully used to discover live hosts on target network using ARP, ICMP, TCP, SYN and UDP- host- discovery techniques.